

Dr. Divyesh Mistry

Computational Materials Scientist

✉ d.mistryg@gmail.com 📞 +34 694290182
🌐 divyesh-mistry.github.io/webpage/ 🔗 linkedin.com/in/dr-divyesh-mistry-7b40a95b/
🐙 github.com/divyesh-mistry 📄 Google Scholar

PROFESSIONAL SUMMARY

Computational Materials Scientist with over a decade of experience driving materials innovation through atomistic simulation, multiscale modeling, and high-performance computing. Proven ability to translate complex physical mechanisms into predictive computational tools for aerospace, defense, and maritime applications. Expertise spans large-scale molecular dynamics (**LAMMPS**), discrete dislocation dynamics, and finite element analysis, underpinned by automated **Python** workflows for data pipelines, defect tracking, and publication-quality visualization on HPC clusters. Research portfolio covers nickel-based superalloys, tungsten alloys, and polymeric adhesive systems, with direct relevance to structural integrity, fracture prediction, and materials qualification.

CORE COMPETENCIES

Simulation & Modeling	⚙️	Molecular Dynamics (MD), Discrete Dislocation Dynamics (DDD), Finite Element Analysis (FEA/FEM), Multiscale Modeling, Fracture Mechanics, Constitutive Modeling, Microstructure Evolution, Defect Mechanics
Simulation Software	🔑	LAMMPS , ABAQUS, ANSYS, COMSOL, OVITO, AtomsK, NEPER, ParaView, MATLAB
Programming & Scientific Python	🔗	Python (NumPy, Pandas, SciPy, Matplotlib), C++, Fortran, Bash, Workflow Automation, Large-Scale Data Analysis, Git/GitHub, L ^A T _E X
High-Performance Computing (HPC)	💻	Linux/Unix, MPI/OpenMP Parallel Computing, HPC Cluster Management, Multi-Million-Atom Simulations, Batch Scheduling

EXPERIENCE

2026 – Present	Postdoctoral Researcher , <i>University of Sevilla, Sevilla, Spain</i> ➤ Build atomistic and multiscale computational models of steel/polymer adhesive interfaces to quantify damage mechanisms, moisture-driven degradation, and joint lifetime under maritime operating conditions. ➤ Develop end-to-end automated Python workflows for LAMMPS simulation management, thermomechanical data processing, and property extraction on HPC clusters. ➤ Bridge atomistic shear response with cohesive-zone and continuum descriptions to support strength prediction and life-assessment models for lightweight hybrid naval structures. ➤ Deliver design tables and maintenance protocols for hybrid structures under U.S. Navy operational conditions. ➤ Project : <i>Multiscale modeling of interfacial delamination of steel/adhesive interfaces by aqueous corrosion</i> — funded by the U.S. Office of Naval Research. ➤ Advisors : Prof. María del Pilar Ariza Moreno and Prof. Michael Ortiz.
April 2026 – July 2026	Visiting Researcher , <i>Merrimack College, Department of Mechanical and Electrical Engineering, North Andover, MA, USA</i> ➤ Conducted large-scale molecular dynamics simulations of temperature-dependent crack-tip deformation and ductile-to-brittle transition behavior in tungsten and tungsten-rhenium alloys. ➤ Developed automated LAMMPS workflows for high-performance computing clusters to examine the effects of temperature, alloy composition, and defect structure on mechanical response. ➤ Built custom Python pipelines using NumPy and Pandas for trajectory processing, stress-strain analysis, defect characterization, and visualization of large atomistic datasets. ➤ Investigated the roles of dislocation activity, crack-tip plasticity, and local structural evolution in controlling stress relaxation and fracture behavior. ➤ Complemented the atomistic analysis with a representative three-dimensional polycrystalline DDD framework for studying Frank–Read source activation, dislocation multiplication, and density-dependent hardening.

Jan 2025 – Dec 2025	Junior Research Fellow (DRDO-Sponsored Project), IIT Bombay, Aerospace Engineering, Mumbai, India (concurrent with final thesis defense and completion) › Characterized dynamic mechanical behavior of SMA-based composites for aerospace and defense applications using Split Hopkinson Pressure Bar (SHPB) experiments. › Calibrated and validated constitutive models by integrating experimental data with numerical simulations, enabling reliable prediction of high-strain-rate structural response. › Delivered data analysis, technical reporting, and mentoring of junior researchers.
Jan 2019 – June 2025	Ph.D. Research Scholar, IIT Bombay, Aerospace Engineering, Mumbai, India › Designed and executed multi-million-atom LAMMPS simulations on HPC clusters to quantify generalized stacking fault energies, dislocation mobility, precipitate shearing, and critical resolved shear stress in powder-metallurgy Ni-based superalloys. › Developed a physics-informed hierarchical multiscale framework mapping atomistic parameters (drag coefficients, obstacle strengths) directly into mesoscale DDD and continuum models — enabling prediction of macroscopic yield and hardening behavior. › Built automated Python data pipelines to parse high-volume MD trajectory data, track defect evolution via OVITO, and generate publication-quality analysis outputs. › Authored a custom parallel 3D polycrystalline DDD code coupled with FEA to model prior particle boundary (PPB) effects, slip localization, and residual stress retention in Ni superalloys. › Published first-author research in top-tier journals, presented at international conferences (EMI, IMPLAST, COMPLAS), and collaborated with experimental teams.
Jan 2019 – Dec 2023	Graduate Teaching Assistant, IIT Bombay, Aerospace Engineering, Mumbai, India › Assisted instruction in Continuum Mechanics, Finite Element Methods (FEM), and Multiscale Modelling for undergraduate and postgraduate engineering courses. › Developed and supervised computational laboratory exercises involving numerical methods, finite element analysis, and scientific programming.
June 2016 – Dec 2018	Assistant Professor, CMR Institute of Technology, Mechanical Engineering, Bangalore, India › Taught undergraduate courses in Finite Element Methods, Experimental Stress Analysis, and Solid Mechanics. › Developed computational laboratory modules using ANSYS, MATLAB, and Python to support engineering analysis and numerical modelling.
June 2014 – July 2016	Graduate Teaching Assistant & M.Tech. Researcher, IIT Kharagpur, Mechanical Engineering, Kharagpur, India › Developed and validated computational models using ABAQUS and ANSYS for structural mechanics and vibration analyses. › Conducted laboratory instruction in Finite Element Modelling, Vibration Analysis, and Tribology, integrating theoretical concepts with numerical simulations.

EDUCATION

Jan 2019 – Dec 2025	 INDIAN INSTITUTE OF TECHNOLOGY - BOMBAY, MUMBAI, INDIA Doctorate, Aerospace Engineering <i>Thesis</i> : Multiscale Modeling of Prior Particle Boundaries in Nickel-based Superalloys <i>CGPA</i> : 9.36/10 <i>Funding</i> : MHRD Teaching Assistantship through Project <i>Key Subjects</i> : Multiscale Modeling, Continuum Mechanics, Plasticity, Computational Mechanics <i>Advisors</i> : Prof. Amuthan A. Ramabathiran, Prof. P. J. Guruprasad
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- Jun 2014 – Jul 2016**  INDIAN INSTITUTE OF TECHNOLOGY - KHARAGPUR, WEST BENGAL, INDIA
M.Tech., Mechanical Engineering (Mechanical System Design)
Thesis : A Mesh-Free Model for Static Analysis of Smart Composite Beams
CGPA : 8.85/10 *Funding* : MHRD GATE Fellowship
Key Subjects : Solid Mechanics, Fracture Mechanics, Finite Element Methods, Composites, Elasticity, Vibration
Advisor : Prof. Manas Chandra Ray
- Dec 2013**  THE AERONAUTICAL SOCIETY OF INDIA, NEW DELHI
B.E., Aeronautical Engineering (Aero Mechanical)
Key Subjects : Aircraft Structures, Aerodynamics, Propulsion, Stability & Control, Solid and Fluid Mechanics

faBook PUBLICATIONS

Peer-Reviewed Journal Articles

- [J1] Mistry, D. A., Tak, T. N., Sankarasubramanian, R., & Guruprasad, P. J.,
A Discrete Dislocation Dynamics Study of Prior Particle Boundary Effects in Ni-based Superalloys,
Journal of Engineering Materials and Technology, pp. 1–55 (2026). Published online : August 4, 2026. DOI :
10.1115/1.4072448.
- [J2] Mistry, D. A. & Ramabathiran, A. A.,
Size-Dependent Power Laws for Edge Dislocations in Nickel Superalloys : A Molecular Dynamics Study,
Computational Materials Science, 259, 114122 (2025). DOI : 10.1016/j.commatsci.2025.114122.

Manuscripts in Preparation

- [P1] Mistry, D. A. & Mahata, A.,
Crack-Tip Plasticity and Ductile-to-Brittle Transition in W and W–Re Systems through Atomistic Fracture and Mesoscale Dislocation Dynamics.
Manuscript in Preparation, 2026.
- [P2] Mistry, D. A., Tak, T. N., & Guruprasad, P. J.,
Polycrystalline Discrete Dislocation Dynamics of Ni-Based Superalloys : Interactions with Prior Particle Boundaries and Second Phases Using Atomistically Informed Inputs.
Manuscript in Preparation.

Conference Contributions

- [C1] Mistry, D. A., Mahata, A., Ramabathiran, A. A., Guruprasad, P. J., & Tak, T. N.,
Multiscale Modeling of Dislocation–Precipitate–PPB Interactions in Powder-Metallurgy Nickel Superalloys.
Engineering Mechanics Institute Conference (EMI 2026), June 2–5, 2026, Boulder, Colorado, USA.
- [C2] Mistry, D. A., Tak, T. N., & Guruprasad, P. J.,
Hierarchical Multiscale Modeling of Plasticity in Ni-Based Superalloys : The Combined Role of Prior Particle Boundaries and Second Phases.
14th International Symposium on Plasticity and Impact Mechanics (IMPLAST 2025), October 12–16, 2025, IIT Roorkee, India.
- [C3] Mistry, D. A., Guruprasad, P. J., & Ramabathiran, A. A.,
An Atomistically Informed Discrete Dislocation Dynamics Study of Prior Particle Boundaries in Ni-Based Superalloys.
17th Conference on Computational Plasticity (COMPLAS), September 5–7, 2023, Barcelona, Spain.

CERTIFICATIONS

- Jan 2020**  Using Python for Research, edX, HarvardX, Harvard University.
- Aug 2020**  The Unix Workbench, Coursera, Johns Hopkins University.
- Aug 2020**  Material Behavior, Coursera, Georgia Institute of Technology.
- May 2018**  A Hands-on Introduction to Engineering Simulations, edX, CornellX, Cornell University.

AWARDS, GRANTS & PROFESSIONAL MEMBERSHIPS

2025		Member , The Minerals, Metals & Materials Society (TMS), USA
2025		Institution of Eminence Travel Grant , IIT Bombay — funded attendance at <i>TMS 2025</i> , USA
2023		IIT Bombay International Travel Grant — funded attendance at <i>COMPLAS 2023</i> , Barcelona, Spain
2019		MHRD Teaching Assistantship (via IRCC, IIT Bombay) — doctoral fellowship award
2014		GATE All India Rank : 69 — Graduate Aptitude Test in Engineering ; MHRD fellowship for master's study
2013		Associate Member , Aeronautical Society of India (AeSI) — Permanent Membership